



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

AUGUST 2006
TRIAL HIGHER SCHOOL
CERTIFICATE
YEAR 12

Mathematics

General Instructions:

- Reading time—5 minutes.
- Working time—3 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 5 sections:
Section A(Questions 1 and 2),
Section B(Questions 3 and 4),
Section C(Questions 5 and 6),
Section D(Questions 7 and 8),
Section E(Questions 9 and 10).

Total marks—120 Marks

- Attempt questions 1–10.
- All questions are of equal value.

Examiner: Mr P.Bigelow

This is an assessment task only and does not necessarily reflect
the content or format of the Higher School Certificate.

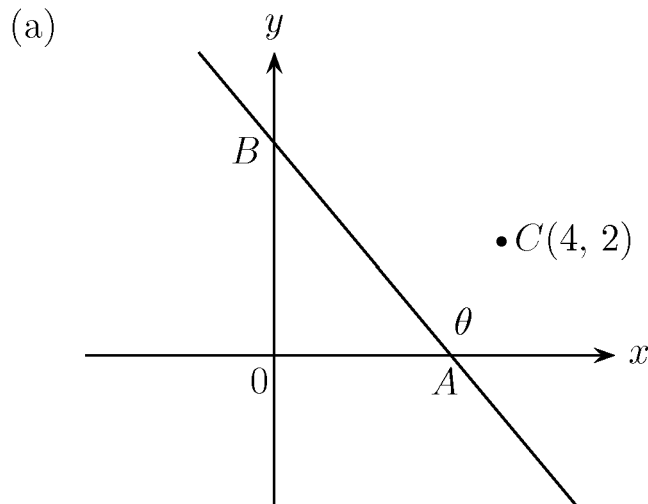
Section A — Start a new booklet

Marks

Question 1 (12 marks)

- (a) Find integers a and b such that $x^2 + 6x + 14 \equiv (x + a)^2 + b$. 2
- (b) Find $e^{2.5}$ correct to 2 decimal places. 2
- (c) What is the exact value of $\cos \frac{7\pi}{6}$? 2
- (d) Solve $|4 - x| = 7$. 2
- (e) By rationalising the denominator, express $\frac{4}{\sqrt{5} - \sqrt{3}}$ in simplest form. 2
- (f) Solve $a^2 = 12a$. 2

Question 2 (12 marks)



The line $4x + 3y - 12 = 0$ has x and y intercepts A and B respectively and makes an angle θ with the positive direction of the x -axis.

C is the point $(4, 2)$.

- (i) Write down the coördinates of points A and B . 2
- (ii) Find the value of θ to the nearest degree. 2
- (iii) Find the perpendicular distance of C from the line $4x + 3y - 12 = 0$. 2
- (iv) Find the area of the triangle ABC . 2
- (b) Solve the pair of simultaneous equations 2

$$\begin{aligned} 3x - y &= 16, \\ x + 4y &= 1. \end{aligned}$$

- (c) Consider the parabola 2

$$y = x^2 - 4x + 8.$$

Find the coördinates of the focus.

Question 3 (12 marks)

- (a) A vessel sails 12 km due north from a port P to A . A second boat sails 20 km from P to B on a bearing of 120° .

(i) What is the distance AB ? 2

(ii) What is the bearing of B from A , correct to the nearest minute? 2

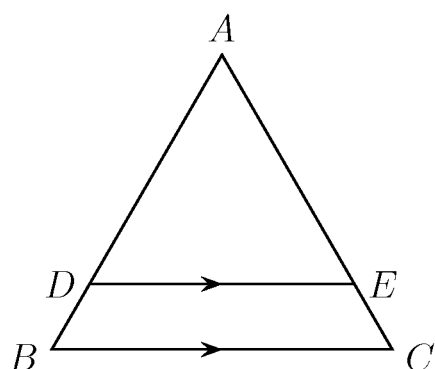
- (b) Differentiate

(i) $\frac{2}{x^4}$ 1

(ii) $\sin(x^3)$ 1

(iii) $x \tan x$ 2

- (c)



In the diagram $DE \parallel BC$. $AB = 16$ cm, $AE = 18$ cm and $EC = 6$ cm.

(i) Prove that $\triangle ADE \sim \triangle ABC$. 2

(ii) Find the length of DB . 2

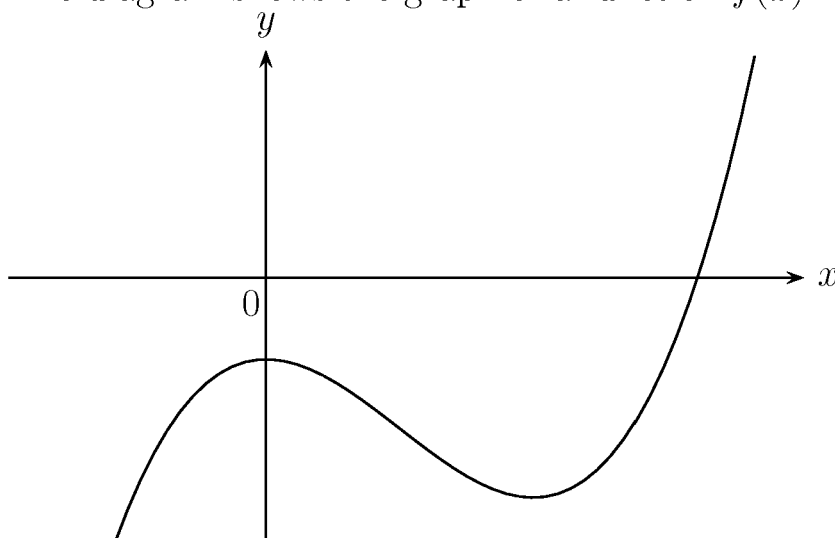
Question 4 (12 marks)

- (a) Evaluate $\int_0^1 \frac{dx}{1+x}$ 2
(leave your answer in exact form).
- (b) Solve $\sqrt{3}\tan x = 1$ for $0 \leq x \leq 2\pi$. 2
- (c) Simplify $\sqrt{\frac{1 - \cos^2 A}{1 + \tan^2 A}}$. 2
- (d) Find the slope of the tangent to the curve $y = \cos\left(x + \frac{\pi}{3}\right)$ at the point $\left(0, \frac{1}{2}\right)$. 2
- (e) Find
- (i) $\int \cos 2x \, dx$ 1
- (ii) $\int \frac{4}{e^{3x}} \, dx$ 1
- (f) Find the values of c for which the equation $x^2 + (c - 2)x + 4 = 0$ has real roots. 2

Question 5 (12 marks)

(a) Write down a quadratic equation with roots $1 + \sqrt{3}$ and $1 - \sqrt{3}$. 2

(b) The diagram shows the graph of a function $f(x)$. 2



(i) Copy this graph.

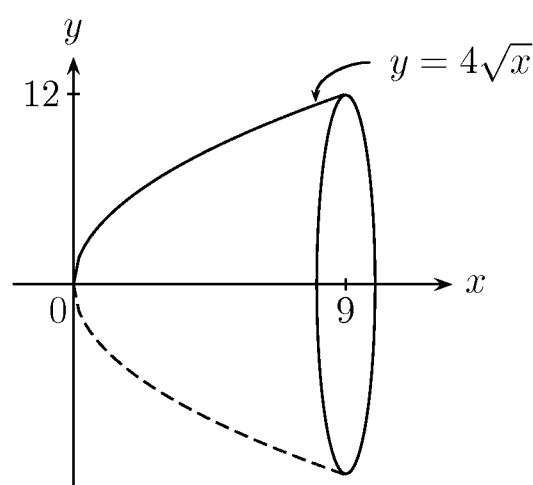
(ii) On the *same* set of axes, draw a sketch of the derivative $f'(x)$ of the function.

(c) The positive multiples of 7 are 7, 14, 21, ...

(i) What is the largest multiple of 7 less than 1200? 2

(ii) What is the sum of the positive multiples of 7 which are less than 1200? 2

(d)



The region enclosed by the curve $y = 4\sqrt{x}$ and the x -axis between $x = 0$ and $x = 9$ is rotated about the x -axis, as shown in the diagram. Find the volume of revolution. 2

(e) The graph of $y = f(x)$ passes through $(2, 5)$ and $f'(x) = 3x^2 + 2$. Find $f(x)$. 2

Question 6 (12 marks)

(a) Given the curve with equation

$$y = x^3 - 3x^2 - 9x + 2.$$

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. 2(ii) Find the coördinates of the stationary points and determine their nature. 2(iii) Sketch the graph of the function for the domain $-2 \leq x \leq 5$. 1(iv) State the maximum value of the function over this domain. 1(b) (i) Copy and then complete the table for $y = \operatorname{cosec} \frac{\pi x}{6}$. 1

x	1	2	3
y			

(ii) Using Simpson's Rule with three function values find an approximate value for 2

$$\int_1^3 \operatorname{cosec} \frac{\pi x}{6} dx.$$

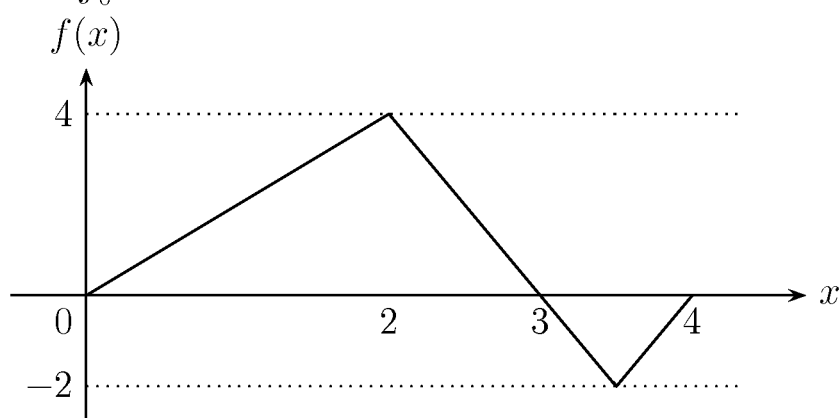
(c) The population of Goldtown is given by $P = 30\,000e^{-0.08t}$.(i) Find the time to the nearest year for the population to halve. 1(ii) Find the decline in the population of Goldtown during the ninth year. 2

Question 7 (12 marks)

- (a) Make a sketch of a continuous curve $y = f(x)$ that has the following properties: [2]
 $f(x)$ is odd, $f(3) = 0$, $f'(1) = 0$.
 $f'(x) > 0$ for $x > 1$,
 $f'(x) < 0$ for $0 \leq x < 1$.

- (b) A bag contains three times as many red marbles as white marbles. [1]
 If a marble is chosen at random, what is the probability that it is white?

- (c) Find $\int_0^4 f(x) dx$ for the following function. [2]



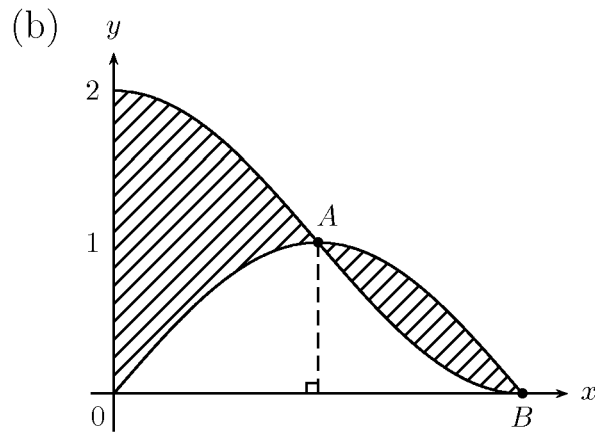
- (d) Simone borrows \$20 000 over 4 years at a rate of 1% compound interest per month. If she pays off the loan in 4 equal yearly instalments find [1]
- (i) the amount she will owe after one month. [1]
 - (ii) the amount she will owe after the first year, just before she pays the first instalment. [1]
 - (iii) the amount of each instalment. [2]
 - (iv) the total amount of interest she will pay. [1]
- (e) Find the limiting sum of the geometric series [2]

$$4 - 2\sqrt{2} + 2 - \dots$$

Question 8 (12 marks)

(a) Evaluate $\int_0^{\ln 4} e^{-2x} dx$.

2



3

The graphs of $y = \sin x$ and $y = 1 + \cos x$ are shown intersecting at $A(\frac{\pi}{2}, 1)$ and $B(\pi, 0)$.

Calculate the total area of the two shaded regions.

- (c) Water is being released from a dam. The rate of flow, F megalitres per hour is given by $F = t(t - 12)^2$, where t is the number of hours since the flow began.

The function applies until the flow ceases.

- (i) For how long does the water flow?
 (ii) Find the maximum rate of flow.
 (iii) What is the total volume of water released?

2

2

3

Question 9 (12 marks)

- (a) The displacement of a particle x metres from the origin, at time t seconds, is given by

$$x = \frac{1}{3}t^3 - 6t^2 + 27t - 18.$$

- (i) Find expressions for velocity and acceleration. 2
- (ii) When is the acceleration zero? 2
- (iii) Where is the particle at this time and what is its velocity? 2
- (b) A uniform cube has three green faces, two white faces, and one red face. If a player throws a green face they win; if red, they lose; and if white they throw again. Robert will throw until he either wins or loses. What is the probability that
- (i) Robert wins with his third throw? 2
- (ii) Robert wins with his first, second, or third throw? 2
- (iii) Robert wins? 2

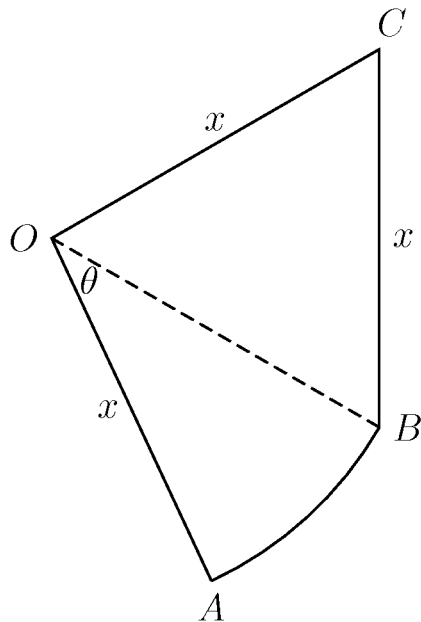
Question 10 (12 marks)

- (a) Solve for x (correct to 3 significant figures)

2

$$3^{x-2} = 50$$

- (b)



The diagram shows a sector OAB of a circle, centre O , and radius x metres. Arc AB subtends an angle θ radians at O . An equilateral triangle BCO adjoins the sector.

- (i) Write down expressions for the

(α) area of sector OAB

1

(β) area of the triangle BCO

1

(γ) length of the arc AB .

1

- (ii) Hence write down expressions for the

(α) area

1

(β) perimeter of the figure $OABC$.

1

- (iii) The perimeter of this figure is $(12 - 2\sqrt{3})$ metres.

(α) For what value of x is the area a maximum?

3

(β) Show that the maximum area is $(6 - \sqrt{3}) \text{ m}^2$.

2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

QUESTION 1

a) $x^2 + 6x + 14$
 $= x + 6x + 9 + 5$

$= (x+3)^2 + 5$

$= (x+a)^2 + b$

$a=3$ & $b=5$

b) $e^{2.5} \div 12.18$ (2 dp)

c) $\cos \frac{7\pi}{6} = -\cos \frac{\pi}{6}$

$= -\frac{\sqrt{3}}{2}$

d) $|4-x|=7$

$4-x=7$ or $4-x=-7$

$x=-3$ or $x=11$

e) $\frac{4}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$

$= \frac{4(\sqrt{5}+\sqrt{3})}{5-3}$

$= 2(\sqrt{5}+\sqrt{3})$

f) $a^2 = 12a$

$a(a-12)=0$

$a=12$ or $a=0$

QUESTION 2

a) i) $4x + 3y - 12 = 0$

when $x=0$ $y=4$

when $y=0$ $x=3$ $A(3,0)$ & $B(0,4)$

ii) $\tan(180-\theta) = \frac{4}{3}$

$180-\theta = \tan^{-1}(\frac{4}{3})$

$180-\theta = 53^\circ 8'$

$\theta = 127^\circ$ nearest degree.

iii) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$C = (x_1, y_1)$
 $= (4, 2)$

$= \frac{|16 + 6 - 12|}{5}$

$= 2$ units

iv) $A = \frac{1}{2}bh$

$= \frac{1}{2} \times \sqrt{4^2 + 3^2} \times 2$

$= 5$ units²

B) $3x - y = 16$ — ①
 $x + 4y = 1$ — ② $\Rightarrow x = 1 - 4y$ ③

Subs ③ into ①

$3(1-4y) - y = 16$
 $-13y = 13$ $y = -1$

Subs $y = -1$ into ②

$x - 4 = 1$ $\therefore$ $x = 5$

c) $y = x^2 - 4x + 8$

$y = (x-2)^2 + 4$

$(x-2)^2 = 4(\frac{1}{4})(y-4)$

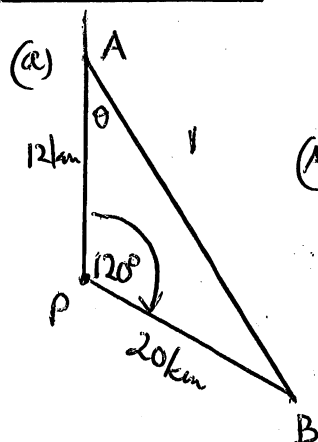
vertex = $(2, 4)$

focal length = $\frac{1}{4}$

$\therefore$ focus = $(2, 4\frac{1}{4})$

SECTION B

Question 3



$$(i) (AB)^2 = 12^2 + 20^2 - 2 \cdot 12 \cdot 20 \cos 120^\circ$$

$$AB = \sqrt{12^2 + 20^2 - 2 \cdot 12 \cdot 20 \cos 120^\circ} = 28 \text{ km}$$

$$(ii) \frac{\sin \theta}{20} = \frac{AB \sin 120^\circ}{AB}$$

$$\theta = 1^\circ 38' 13''$$

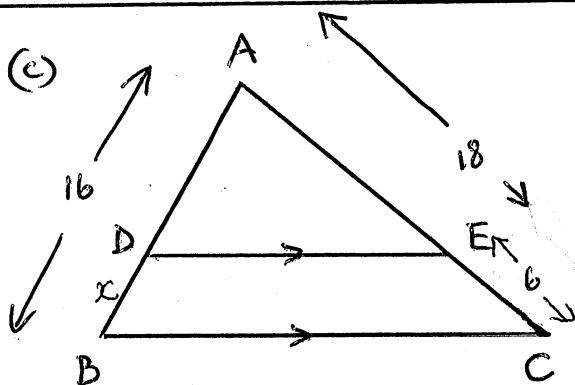
$\therefore$ Bearing is

$$(180 - \theta) \text{ i.e. } 141^\circ 47'$$

$$(b) (i) y = 2x^{-4} \Rightarrow y' = -8x^{-5}$$

$$(ii) y = \sin(x^3) \Rightarrow y' = 3x^2 \cos x^3$$

$$(iii) y = x \tan x \Rightarrow y' = x \sec^2 x + 1 \cdot \tan x$$



$$(i) \angle A \text{ is common}$$

$$\angle ADE = \angle ABC \text{ (corresp. angles on } \parallel \text{ lines)}$$

$\therefore$ Triangles ABC and ADE are equiangular $\Rightarrow$ Similar.

$$(ii) \text{ let } BD = x \Rightarrow AD = 16 - x$$

$$\frac{16-x}{16} = \frac{18}{24} \Rightarrow \boxed{x=4}$$

$$4x = 16$$

Question 4

$$(a) \left[\log_e(1+x) \right]_0^1 = \log_e 2 - \log_e 1$$

$$= \log_e 2$$

$$(b) \sqrt{3} \tan x = 1$$

$$\Rightarrow \tan x = \frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{6} \text{ and } \pi + \frac{\pi}{6}$$

$$\text{i.e. } x = \frac{\pi}{6}, \frac{7\pi}{6}$$

$$(c) \sqrt{\frac{1 - \cos^2 A}{1 + \tan^2 A}} = \sqrt{\frac{\sin^2 A}{\sec^2 A}}$$

$$= \sqrt{\sin^2 A \cos^2 A}$$

$$= \sin A \cos A$$

$$(d) y = \cos(x + \frac{\pi}{3})$$

$$y' = -\sin(x + \frac{\pi}{3})$$

$$\text{At } x=0 \text{ grad. tang. is } -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$$

$$(e) (i) \int \cos 2x \, dx = \frac{1}{2} \sin 2x + c$$

$$(ii) \int \frac{4}{e^{3x}} \, dx = 4 \int e^{-3x} \, dx$$

$$= -\frac{4}{3} e^{-3x} + c$$

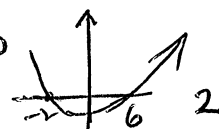
$$(f) x^2 + (c-2)x + 4 = 0$$

For real roots $\Delta \geq 0$

$$\Delta = (c-2)^2 - 4(1)(4)$$

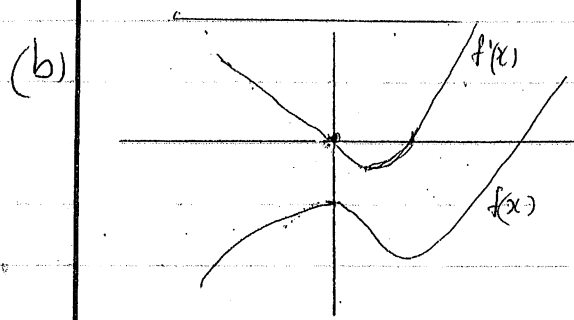
$$= (c-6)(c+2)$$

$$\geq 0 \text{ when } \boxed{c \leq -2 \text{ or } c \geq 6}$$



QUESTION 5

(a) $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
 $x^2 - x(1 - \sqrt{3} + 1 + \sqrt{3}) + (1 + \sqrt{3})(1 - \sqrt{3})$
 $x^2 - 2x - 2 = 0$



(c) $7 + 7(n-1) < 1200$
 (i) $7 + 7n - 7 < 1200$
 $n < 171.4$
 $n = 171$

multiple = 1197

(ii) $\frac{171}{2} (7 + 1197)$
 $\text{Sum} = 102942$

(d) $V = \pi \int_0^9 (4\sqrt{x})^2 dx$
 $= \pi \int_0^9 16x dx$
 $= \pi [8x^2]_0^9$
 $= 648\pi \text{ u}^3$

(e) $f'(x) = 3x^2 + 2$
 $f(x) = x^3 + 2x + c$
 (2, 5) $5 = 8 + 4 + c, c = -7$
 $f(x) = x^3 + 2x - 7$

QUESTION 6

$y = x^3 - 3x^2 - 9x + 2$

(i) $\frac{dy}{dx} = 3x^2 - 6x - 9$

$\frac{d^2y}{dx^2} = 6x - 6$

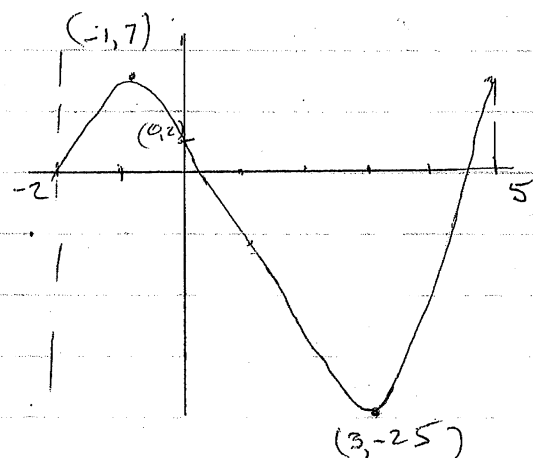
stat pts $3x^2 - 6x - 9 = 0$

$(x - 3)(x + 1) = 0$

(ii) pts $(3, -25)$ and $(-1, 7)$

$x = 3 \quad \frac{d^2y}{dx^2} > 0 \therefore \text{min}$

$x = -1 \quad \frac{d^2y}{dx^2} < 0 \therefore \text{max}$



(iv) max = 7 at $x = -1, 5$

(b)

x	1	2	3
y	2	$\frac{2}{\sqrt{3}}$	1

 $y = \operatorname{Cosec}\left(\frac{\pi x}{6}\right)$

$A \approx \frac{1}{3} \left(1 + 2 + 4 \times \frac{2}{\sqrt{3}}\right)$
 $= 2.54$

(c) $15000 = 30000e^{-0.08t}$

(i) $\frac{1}{2} = e^{-0.08t}$

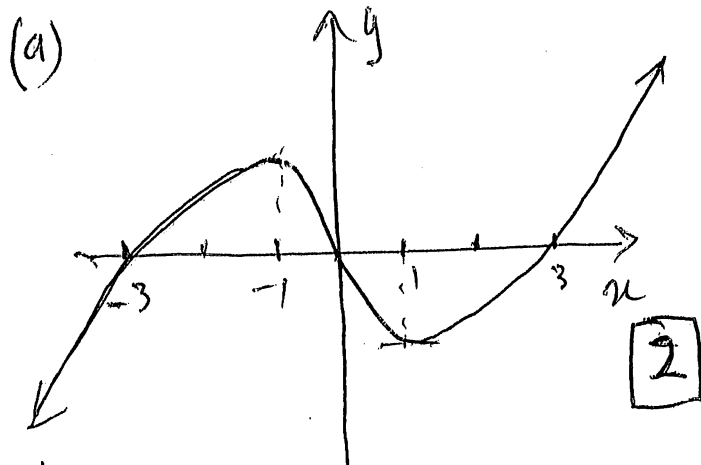
$\ln\left(\frac{1}{2}\right) = -0.08t$

$t = -\frac{\ln \frac{1}{2}}{0.08} = 8.66 \rightarrow 9 \text{ years}$

(ii) $30000(e^{-0.08 \times 8} - e^{-0.08 \times 9})$
 $= 1216$

decline 1216 people

Question 7



(b) $\frac{1}{4}$ [1]

(c) $\int_0^4 f(x) dx = \frac{1}{2} \times 3 \times 4 - \frac{1}{2} \times 1 \times 2$
 $= 6 - 1$
 $= 5$ [2]

(d) Let $P = \$20000$
 $R = 1.01$
 $A_n = \text{Amount owing after } n \text{ months}$
 $Q = \text{annual instalment.}$

(i) $A_1 = PR$
 $= \$20000 \times 1.01$
 $= \$20200$ [1]

(ii) $A_{12} = PR^{12}$
 $= \$20000 \times 1.01^{12}$
 $= \$22536.50$ [1]

(iii) $A_{13} = (PR^{12} - Q)R$
 $A_{36} = PR^{36} - QR^{24} - QR^{12}$
 $A_{48} = PR^{48} - QR^{36} - QR^{24} - QR^{12}$
 $= PR^{48} - \frac{Q(R^{48} - 1)}{R^{12} - 1}$

But $A_{48} = Q$

$$\therefore Q = \frac{PR^{48}(R^{12} - 1)}{R^{48} - 1}$$

$$= \frac{\$20000(1.01)^{48}(1.01^{12} - 1)}{1.01^{48} - 1}$$

$$= \$6679.59 \quad [2]$$

(iv) Total Interest
 $= 4Q - \$20000$
 $= \$6718.36$ [1]

(e) GS: $4 - 2\sqrt{2} + 2 - \dots$
 $a = 4 \quad r = \frac{-2\sqrt{2}}{4}$
 $= -\frac{1}{\sqrt{2}}$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{4}{1 + \frac{1}{\sqrt{2}}}$$

$$= \frac{4\sqrt{2}}{\sqrt{2} + 1}$$

$$= 8 - 4\sqrt{2}$$

(≈ 2.343)

($= \frac{8}{2+\sqrt{2}}$)

[2]

Question 8

(a) $f(x) = 2 - x^2$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 - (x+h)^2 - (2 - x^2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 - x^2 - 2xh - h^2 - 2 + x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(-2x - h)}{h}$$

$$= -2x$$

When $x = 1$

$$f'(1) = -2 \quad [2]$$

(b) Area = $\int_0^{\pi/2} ((1 + \cos x) - \sin x) dx$
 $+ \int_{\pi/2}^{\pi} (\sin x - (1 + \cos x)) dx$

$$= \left[x + \sin x + \cos x \right]_0^{\pi/2} + \left[-\cos x - (x + \sin x) \right]_{\pi/2}^{\pi}$$

$$= \left[\left(\frac{\pi}{2} + 1 + 0 \right) - (0 + 0 + 1) \right] + \left[(-(-1) - (\pi + 0)) - (-0 - (\frac{\pi}{2} + 1)) \right]$$

$$= \frac{\pi}{2} + (1 - \pi - (-\frac{\pi}{2} - 1))$$

$$= 2 \text{ unit}^2 \quad [3]$$

(c) $F = t(t-12)^2$

$$F' = t \cdot 2(t-12) + (t-12)^2 \cdot 1$$

$$= (3t-12)(t-12)$$

$$F' = 0 \text{ for } t = 4 \text{ or } 12$$

(i) $F = 0$ when $t = 0, 12, 12$

$\therefore$ Flows for 12 hours $[2]$

(ii) Stationary points at $t = 4$ or 12

$$F'' = 6t - 48$$

$$F''(4) = -24 \quad F''(12) = 24$$

$\therefore$ Rel Max $\quad \quad \quad$ Rel Min

$\therefore$ Max flow when $t = 4$

$$F(4) = 256 \text{ ML/hr} \quad [2]$$

(iii) Total flow

$$= \int_0^{12} (t^3 - 24t^2 + 144t) dt$$

$$= \left[\frac{t^4}{4} - \frac{24t^3}{3} + \frac{144t^2}{2} \right]_0^{12}$$

$$= \left[\frac{t^4}{4} - 8t^3 + 72t^2 \right]_0^{12}$$

$$= 1728 \text{ ML} \quad [3]$$

Q 9(a) $x = \frac{1}{3}t^3 - 6t^2 + 27t - 18$

(i) $\dot{x} = t^2 - 12t + 27$

$\ddot{x} = 2t - 12$ [2]

(ii) $\ddot{x} = 0$ when $t = 6$ s [2]

(iii) When $t=6$, $x = 0$ m
 $\dot{x} = -9$ m/s [2]

(b) (i) $P(WW4) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{2}$
 $= \frac{1}{18}$ [2]

(ii) $P(4) + P(W4) + P(WW4)$
 $= \frac{1}{2} + \frac{1}{3} \times \frac{1}{2} + \frac{1}{18}$
 $= \frac{13}{18}$ [2]

(iii) $P(\text{win}) = P(4) + P(W4) + P(WW4)$
 $+ P(WWW4) + \dots$
 $= \frac{1}{2} + \frac{1}{6} + \frac{1}{18} + \dots$
 $= \frac{(\frac{1}{2})}{(1 - \frac{1}{3})}$
 $= \frac{1}{2} \times \frac{3}{2}$
 $= \frac{3}{4}$ [2]

Q 10 (a) $3^{x-2} = 50$
 $\ln(3^{x-2}) = \ln 50$
 $(x-2) \ln 3 = \ln 50$
 $x-2 = \frac{\ln 50}{\ln 3}$
 $x = 2 + \frac{\ln 50}{\ln 3}$
 $= 5.560876 \dots$
 ≈ 5.56 [2]

(b) (i) (a) $A_{\text{sector}} = \frac{1}{2} x^2 \theta$ m² [1]

(b) $A_{\Delta} = \frac{1}{2} x^2 \cdot \sin \frac{\pi}{3}$
 $= \frac{1}{2} x^2 \cdot \frac{\sqrt{3}}{2}$
 $= \frac{\sqrt{3} x^2}{4}$ m² [1]

(c) $L_{\text{arc}} = x\theta$ [1]

(ii) (a) $A = \frac{1}{2} x^2 \theta + \frac{\sqrt{3} x^2}{4}$
 $= \frac{x^2}{4} (2\theta + \sqrt{3})$ [1]

(b) $P = 3x + x\theta$

$= x(3 + \theta)$ [1]

(iii) $12 - 2\sqrt{3} = x(3 + \theta)$

$\therefore 3 + \theta = \frac{12 - 2\sqrt{3}}{x}$

$\theta = \frac{12 - 2\sqrt{3}}{x} - 3$

$A = \frac{x^2}{4} (2\theta + \sqrt{3})$
 $= \frac{x^2}{4} (2 \times (\frac{12 - 2\sqrt{3}}{x} - 3) + \sqrt{3})$
 $= \frac{x^2}{4} (\frac{24 - 4\sqrt{3}}{x} - 6 + \sqrt{3})$

$= (6 - \sqrt{3})x - (6 - \sqrt{3}) \frac{x^2}{4}$

$= (6 - \sqrt{3})(x - \frac{x^2}{4})$

$A' = (6 - \sqrt{3})(1 - \frac{x}{2})$

$A'' = (6 - \sqrt{3}) \times -\frac{1}{2} < 0$ [3]

$\therefore$ Max area when $x = 2$

Max area $= (6 - \sqrt{3})(2 - 1)$
 $= 6 - \sqrt{3}$ m² [2]